

No. of Functional Features Included in the Solution

Date	08 July 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Input Form	HTML/CSS interface allowing users to input the 4 socio-economic metrics.	Frontend (UI)	Done	High
2	Input Validation	Ensures entered values are within logical bounds (e.g., GNI > 0) before processing.	Frontend / Backend	Done	Medium
3	Data Normalization	Processes and scales the raw user inputs to match the ML training data format.	Backend Logic	Done	High
4	HDI Prediction Engine	Executes the Scikit-learn Linear Regression model to calculate the final HDI score.	Machine Learning	Done	High
5	Results Dashboard	Dynamically displays the predicted HDI score and categorizes the development tier.	Frontend (UI)	Done	High
6	REST API Endpoint	Flask routing (/predict) that receives JSON data and returns predictions.	Backend (API)	Done	Medium
7	Cloud Deployment	Live hosting of the web application making it accessible via a public URL.	Infrastructure	Done	High

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	7
Core / Must-Have Features	5 (Input Form, Normalization, Prediction Engine, Dashboard, API)
Additional / Nice-to-Have Features	3 (Input Validation, Cloud Deployment, Export Results)
Features Tested & Verified	7

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	User Input Form, Results Dashboard
2	Backend / Logic	3	Data Normalization, HDI Prediction Engine, Input Validation
3	Database / Storage	1	Model Serialization (Loading hdi_model.pkl)
4	API / Integration	1	Flask REST API Endpoint (/predict)
5	Security / Authentication	1	Input Data Sanitization / Type Checking